PROCUREMENT PACKAGE:			EL008 - 415V MCCs		
EQUIPMENT NUMBER(S):		EQUIPMENT NAME(S):	CONSTRUCTION	BUSBAR RATING	FAULT WITHSTAND CURRENT RATING
1	F12-SW-001	Primary Crushing LV MCC	Double Sided	3600A	65kA for 1 second
2	F13-SW-002	SAG Mill LV MCC	Double Sided	3600A	65kA for 1 second
3	F13-SW-003	Ball Mill LV MCC	Double Sided	3600A	65kA for 1 second
4	F13-SW-014	Recycle Crushing LV MCC	Double Sided	3600A	65kA for 1 second
5	F13-SW-020	Mill Chillers LV MCC	Double Sided	3600A	65kA for 1 second
6	F16-SW-004	Flotation LV MCC	Double Sided	3600A	65kA for 1 second
7	F16-SW-005	Flotation Blowers and Compressors LV MCC	Double Sided	3600A	65kA for 1 second
8	F16-SW-008	Flotation Emergency Backup LV MCC	Single Sided	960A	65kA for 1 second
9	F21-SW-006	CIL4 LV MCC 1 (SP2)	Double Sided	3600A	65kA for 1 second
10	F21-SW-007	CIL4 LV MCC 2 (SP2)	Double Sided	3600A	65kA for 1 second
11	F22-SW-030	Carbon Treatment & Elution LV MCC 1 (SP2)	Double Sided	3600A	65kA for 1 second
12	F22-SW-031	Carbon Treatment & Elution LV MCC 2 (SP2)	Double Sided	3600A	65kA for 1 second
13	F23-SW-023	Final Tails Pumps LV MCC	Double Sided	1600A	65kA for 1 second
14	F28-SW-009	Ultrafine Grinding LV MCC	Double Sided	1600A	65kA for 1 second
15	F72-SW-072	Reagents LV MCC	Double Sided	2500A	65kA for 1 second

REFERENCE DOCUMENTS:		
STAGE	DRAWING NUMBER	DESCRIPTION
Stage 1 /2	2000-F00-LST-EL-11003	415V MCC Module list
Stage 1	2000-F12-SLD-EL-10101	Primary Crushing, F12-SW-001 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F12-SLD-EL-10102	Primary Crushing, F12-SW-001 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F12-SLD-EL-10103	Primary Crushing, F12-SW-001 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F12-SLD-EL-10103	Primary Crushing, F12-SW-001 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F13-SLD-EL-10301	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F13-SLD-EL-10302	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F13-SLD-EL-10303	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F13-SLD-EL-10304	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F13-SLD-EL-10305	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 5
Stage 1	2000-F13-SLD-EL-10306	Milling & Classification – SAG Mill, F13-SW-002 415V MCC, Single Line Diagram, Sheet 6
Stage 1	2000-F13-SLD-EL-10901	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F13-SLD-EL-10902	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F13-SLD-EL-10903	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F13-SLD-EL-10904	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F13-SLD-EL-10905	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 5
Stage 1	2000-F13-SLD-EL-10906	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 6
Stage 1	2000-F13-SLD-EL-10907	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 7
Stage 1	2000-F13-SLD-EL-10908	Milling & Classification – Ball Mill, F13-SW-003 415V MCC, Single Line Diagram, Sheet 8

REFERENCE DOCUMENTS:		
STAGE	**DRAWING NUMBER**	**DESCRIPTION**
Stage 1	2000-F13-SLD-EL-10401	Recycle Crushing, F13-SW-014 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F13-SLD-EL-10402	Recycle Crushing, F13-SW-014 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F13-SLD-EL-10403	Recycle Crushing, F13-SW-014 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F13-SLD-EL-10404	Recycle Crushing, F13-SW-014 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F13-SLD-EL-10701	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F13-SLD-EL-10702	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F13-SLD-EL-10703	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F13-SLD-EL-10704	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F13-SLD-EL-10705	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 5
Stage 1	2000-F13-SLD-EL-10706	Mill Chillers and Compressors, F13-SW-020 415V MCC, Single Line Diagram, Sheet 6
Stage 1	2000-F16-SLD-EL-10201	Flotation and Pre-Leach Thickening, F16-SW-004 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F16-SLD-EL-10202	Flotation and Pre-Leach Thickening, F16-SW-004 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F16-SLD-EL-10203	Flotation and Pre-Leach Thickening, F16-SW-004 415V MCC, Single Line Diagram, Sheet 3
Stage 1	2000-F16-SLD-EL-10204	Flotation and Pre-Leach Thickening, F16-SW-004 415V MCC, Single Line Diagram, Sheet 4
Stage 1	2000-F16-SLD-EL-10205	Flotation and Pre-Leach Thickening, F16-SW-004 415V MCC, Single Line Diagram, Sheet 5
Stage 1	2000-F16-SLD-EL-10211	Rougher and Scavenger Flotation, F16-SW-005 415V MCC, Single Line Diagram, Sheet 1
Stage 1	2000-F16-SLD-EL-10212	Rougher and Scavenger Flotation, F16-SW-005 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F16-SLD-EL-10221	Flotation Emergency Backup, F16-SW-008 415V MCC, Single Line Diagram
Stage 2	2000-F21-SLD-EL-10201	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 1
Stage 2	2000-F21-SLD-EL-10202	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 2
Stage 2	2000-F21-SLD-EL-10203	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 3
Stage 2	2000-F21-SLD-EL-10204	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 4
Stage 2	2000-F21-SLD-EL-10205	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 5
Stage 2	2000-F21-SLD-EL-10206	Flotation Tailings CIL4, F21-SW-006 415V MCC, Single Line Diagram, Sheet 6
Stage 2	2000-F21-SLD-EL-10251	Flotation Tailings CIL4, F21-SW-007 415V MCC, Single Line Diagram, Sheet 1
Stage 2	2000-F21-SLD-EL-10252	Flotation Tailings CIL4, F21-SW-007 415V MCC, Single Line Diagram, Sheet 2
Stage 2	2000-F21-SLD-EL-10253	Flotation Tailings CIL4, F21-SW-007 415V MCC, Single Line Diagram, Sheet 3
Stage 2	2000-F21-SLD-EL-10254	Flotation Tailings CIL4, F21-SW-007 415V MCC, Single Line Diagram, Sheet 4
Stage 2	2000-F21-SLD-EL-10255	Flotation Tailings CIL4, F21-SW-007 415V MCC, Single Line Diagram, Sheet 5
Stage 2	2000-F22-SLD-EL-10201	Carbon Treatment & Elution, F22-SW-030 415V MCC, Single Line Diagram, Sheet 1
Stage 2	2000-F22-SLD-EL-10202	Carbon Treatment & Elution, F22-SW-030 415V MCC, Single Line Diagram, Sheet 2
~~Stage 2~~	~~2000-F22-SLD-EL-10203~~	~~Carbon Treatment & Elution, F22-SW-030 415V MCC, Single Line Diagram, Sheet 3~~
~~Stage 2~~	~~2000-F22-SLD-EL-10204~~	~~Carbon Treatment & Elution, F22-SW-030 415V MCC, Single Line Diagram, Sheet 4~~
Stage 2	2000-F22-SLD-EL-10251	Carbon Treatment & Elution, F22-SW-031 415V MCC, Single Line Diagram, Sheet 1
Stage 2	2000-F22-SLD-EL-10252	Carbon Treatment & Elution, F22-SW-031 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F23-SLD-EL-10201	Final Tailings Handling & Storage, F23-SW-023 415V MCC, Single Line Diagram, Sheet 1

REFERENCE DOCUMENTS:		
STAGE	**DRAWING NUMBER**	**DESCRIPTION**
Stage 1	2000-F23-SLD-EL-10202	Final Tailings Handling & Storage, F23-SW-023 415V MCC, Single Line Diagram, Sheet 2
Stage 1	2000-F28-SLD-EL-10101	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 1
Stage 1	2000-F28-SLD-EL-10102	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 2
Stage 1	2000-F28-SLD-EL-10103	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 3
Stage 1	2000-F28-SLD-EL-10104	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 4
Stage 1	2000-F28-SLD-EL-10105	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 5
Stage 1	2000-F28-SLD-EL-10106	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 6
Stage 1	2000-F28-SLD-EL-10107	415V F28-SW-009 Ultrafines Single Line Diagram Sheet 7
Stage 2	2000-F72-SLD-EL-10201	415V F72-SW-072 Reagents Single Line Diagram Sheet 1
Stage 2	2000-F72-SLD-EL-10202	415V F72-SW-072 Reagents Single Line Diagram Sheet 2
Stage 2	2000-F72-SLD-EL-10203	415V F72-SW-072 Reagents Single Line Diagram Sheet 3
Standard drawing	2000-F00-STD-EL-10100	415V Fxx-SW-xxx Incomer Schematic Diagram Sheet 1
Standard drawing	2000-F00-STD-EL-10101	415V Fxx-SW-xxx Incomer Schematic Diagram Sheet 2
Standard drawing	2000-F00-STD-EL-10102	415V Fxx-SW-xxx 24VDC Distribution Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10103	415V Fxx-SW-xxx 0.37..30kW DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10104	415V Fxx-SW-xxx 37..75kW DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10105	415V Fxx-SW-xxx 90..220kW DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10106	415V Fxx-SW-xxx 250kW+ DOL Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10107	415V Fxx-SW-xxx 0.37..30kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10108	415V Fxx-SW-xxx 37..75kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10109	415V Fxx-SW-xxx 90..200kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10110	415V Fxx-SW-xxx 250kW+ VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10111	415V Fxx-SW-xxx 0.37..30kW DOL Sump Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10112	415V Fxx-SW-xxx 0..37..30kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10113	415V Fxx-SW-xxx 37..75kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10114	415V Fxx-SW-xxx 90..220kW DOL Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10115	415V Fxx-SW-xxx 0.37..30kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10116	415V Fxx-SW-xxx 37..75kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10117	415V Fxx-SW-xxx 90..200kW VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10118	415V Fxx-SW-xxx 250kW+ VSD Conveyor Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10119	415V Fxx-SW-xxx 0.37..30kW Reversing DOL Drive Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10120	415V Fxx-SW-xxx 0.37..30kW DOL Dual Motor Screen Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10121	415V Fxx-SW-xxx 0.37..30kW VSD Dual Motor Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10122	415V Fxx-SW-xxx 30..300A Active Harmonic Filter Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10124	415V Fxx-SW-xxx 20..250A Switchroom DB Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10125	415V Fxx-SW-xxx 20..800A MCCB Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10126	415V Fxx-SW-xxx 20..400A Decontactor Feeder Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10127	415V Fxx-SW-xxx DOL Heater Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10128	415V Fxx-SW-xxx 220..250kW VSD Pump Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10129	415V Fxx-SW-xxx 90..220kW Reversing DOL Drive Schematic and Termination Diagram
Standard drawing	2000-F00-STD-EL-10130	415V Fxx-SW-xxx 37..75kW DOL Sump Pump Schematic and Termination Diagram

REFERENCE SPECIFICATIONS:

1. 2000-F00-STS-EL-10014_2 - Preferred Equipment Suppliers For Electrical, Instrumentation And Controls Specification
2. 2000-F00-STS-EL-10016_1 - Low Voltage Switchboards & Switchgear Electrical Specification
3. 2000-F00-STS-GE-10002_1 - Site Specific Information
4. 2000-F00-STS-GE-10003_3 - Plant Numbering Specification
5. 2000-F00-STS-ST-10001_3 - Steelwork Fabrication
6. 2000-F00-STS-ST-10002_2 - Steelwork Erection Specification
7. 2000-MEP-EL008-SCH1 - Vendor Document Schedule
8. SPEC-00-G-05_12 – Technical Specification for Protective Coatings

SCOPE OF WORK

Work Included

- Design, supply and fabrication of fifteen (15) indoor motor control centres including, but not limited to:
  - Withdrawable Form 4a Incomer ACB/s, or fixed Moulded Case Circuit Breaker (MCCB) incomer where bus rating is less than 1000A;
  - Demountable Form 4a Outgoing Modules;
  - All internal MCC Components including circuit breakers, E/L circuit breakers, contactors, electronic overloads (EOLs), thermal overloads (TOLs), soft starters, relays, transformers, UPS 240VAC to 24VDC Power supplies and Distribution, terminals, shrouds etc;
  - Installation of free issue items into top of relevant tiers
    - Cisco network switches installed in the top of tier cable zones to marshal intelligent electronic devices (IEDs) such as EOLs, soft starters and relays. Maximum of 1 per two tiers, vendor to confirm required quantity of ports.
  - CAT6a wiring to IEDs (EOLs, soft starters, relays, etc).
  - Arc fault detection at incomer and all outgoing modules including cable ways. The arc fault detection for the incomer shall be current supervised to trip the incomer and upstream feeder using both current and optical sensors.
  - All internal wiring and cabling including power, control, copper communications and fibre communications;
  - Labelling, including stainless steel equipment nameplate;
  - Lifting lugs, as applicable;
  - Any other items which are normally supplied with this equipment;
- Surface Protection;
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly;
- Factory Acceptance Testing at manufacturer's premises;
- Supply of all necessary documentation, drawings and test documentation;
- Supplier shall provide a schedule of Take In and Take Out pricing for each size of MCC module and for a whole tier;

Special Notes:

- The LV MCC package will be awarded together with the switchroom package EL01A to the successful tenderer.

Work Excluded

- Cabling external to switchboard and switchroom;
- Spare parts;
- Installation and commissioning at site.

Design Notes

EPC·S document package includes MCC Modules List and typical schematics. Design has been described using typical schematics. Actual schematics will be produced shortly following award of contract, if successful. 24V DC distribution schematic will be updated to suit MCC GA following receipt of successful vendor's approved MCC GA drawings.

MCC with bus bar rating <1000A shall be provided with a motorised Moulded Case Circuit Breaker (MCCB) style incomer.

Supplier shall nominate suitable circuit breaker sizes to suit the type 2 high efficiency coordination tables for the offered switchgear. Multi-motor drives shall ensure that electronic overloads and their associated contactors are type 2 high efficiency coordinated to suit the individual motors and shall be suitable for the full load current, noting that vibrating motors (such as those used in screens) typically have a larger full load current than a standard motor.

DOL motor starter modules will utilise electronic overloads with Ethernet communications functionality. Protection functions required from the overload are stated further in this datasheet. Soft Starters, where utilised, will have ethernet communications. IO Expansion modules may be utilised where required (E.g. multi-motor DOL).

MCC incomer arc flash detection scheme protection CT's are to be mounted on the upstream transformer by Vendor.

VSD feeder modules to be fitted with current limiting fuses where it is required to achieve adequate protection to the cable and/or the VSD.

24V DC distribution schematic describes indicative 24V DC distribution throughout the MCC. It is envisaged that an MCB will be installed at the top of each cable zone. 24V DC control supply will be looped between DC distribution compartment and MCBs at top of tier. Within each tier, 24V DC control will loop from the MCB at top of tier to each individual starter module. Supplier shall ensure that their offer includes all necessary equipment for the 24V DC distribution.

Live line indication and earth leakage shall be provided for all incomers and MCC modules. Other forms of imbalance or earth leakage protection required for all hardwired sub-circuits.

Arc fault detection shall be provided at incomer and all outgoing modules including cable ways via optical sensors. The arc fault detection for the incomer shall be current supervised to trip the incomer and upstream feeder using both current and optical sensors. Arc fault detection current transformers to be installed in the transformer room by the Switchroom Vendor.

DUTY SPECIFICATION	**UOM**	EPC **DATA** VTA = Vendor to Advise	**SUPPLIER DATA** Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	**NOTES**
General				
Equipment Number	-	Refer above		1
Equipment Name	-	Refer above		1
Make	-		POWINS	2
Model	-		ECO LOCK PREMIUM	2
MCC Type	-	Air Insulated, Metal Clad, Form 4a, Free Standing, Double Sided	Confirmed	1
ACB Functional Unit Connections	-	Withdrawable	Confirmed	1
Non-ACB Functional Unit Connections (AS61439)	-	Demountable	Confirmed	1
Country of Manufacture	-		Malaysia/Perth Australia	2
Compliance with Australian Standards	Yes/No	Yes – AS3439, AS3947, AS60947, AS61439	Confirmed	1, 6
Drive and Load Descriptions	-	As per MCC Module List – 2000-F00-LST-EL-11003	Confirmed	1
Environmental Conditions				
Plant Location	-	Western Australia		1
Service Location	Indoor/Outdoor	Indoor		1
Ambient Indoor Air Temperature - Min	°C	21		1
Ambient Indoor Air Temperature - Max	°C	27		1
Humidity	%	0 to 99		1
Altitude	m	<1000m		1
Supply				
Supply	-	3P + N + E		1
Input Voltage	V	415V +/- 10%		1
Frequency	Hz	50 (±2%)		1
Neutral Earthing	-	Solid – M.E.N. on MCC (TN-S)		1
Switchboard Ratings				
Type Tested Assembly to AS61439	No/Partially/Fully	Fully Type Tested Assembly	Confirmed	1, 6

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Switchboard Rating Plate	-	Yes	Confirmed	1
Main Bus Bar Continuous Current Rating	A	Refer to Table above. Bus shall be fully rated along its length. Supplier to confirm ratings for site conditions.	Confirmed	3
Neutral Bus Bar Continuous Current Rating	A	100% of phase rating	Confirmed	3
Main & Neutral Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3
Dropper Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3
Earth Bus Bars Fault Withstand Level	kA for sec	VTA	48kA rms at 240V	2
Rated Insulation Voltage	V	1000V	Confirmed	3
Interruption Capacity (All circuits connected directly and indirectly to main bus Including ACBs, MCCB and 240V control circuit protection). ** Cascading is not permitted, fault current limiting fuses to be used where required.	kA	65kA Icu Vendor to Advise kA Ics (ACB incomer shall be rated to 65kA Icu/Ics)	65kA	3
Incomer Current Rating (de-rated according to manufacturer's de-rating tables and enclosure)	A	Refer to table above	Terasaki 4000A ACB derated to 3600A as per type tests	3
Internal Arc Fault Containment (IAC) Rating to IEC TR 61641 / AS 61439.1 Annex ZD - Incomer Line Side Terminals - Main & Dropper Bus Bar System - Outgoing Modules	kA for sec	Supplier to confirm ratings below: 1. 63kA/0.3s minimum 2. IAC rated bus not required if insulated 3. 63kA/0.3s minimum	Confirmed. Outgoing modules to AS61439.1 Annex ZD	3
Cascading used to achieve minimum breaking capacity	Yes / No	No		1
Switchboard Construction				
General Construction	-	Free Standing Metal Clad Air Insulated	Confirmed	3
Plinth Construction	-	VTA	HDG Steel	2
Degree of Ingress Protection	IP	VTA – Min IP42	Confirmed	3
Type of Cooling	-	Air cooled (arc contained non ventilated)	Confirmed	1
Exterior Paint Colour	-	AS2700 Tropical Blue T11	Confirmed	1
Interior Paint Colour	-	AS2700 / White	Confirmed	1
Paint Specification	-	Vendor Standard	Confirmed	2
Single Sided / Double Sided	Single / Double	Double Sided; with exception of F16-SW-008, which is single sided	Confirmed	1
Spare Space Allowance for Future Outgoing Modules		One spare double-sided functional tier and cable zone tier with no module installed on the end the MCC	Confirmed	3
Capacity for Future Extension	Yes / No	Yes, at one end	Confirmed	1
Switchboard Incomer Form	-	Form 4a	Confirmed	1
Switchboard Outgoing Motor Starters and Feeders Form	-	Form 4a	Confirmed	1
Cable Entry	Top / Bottom	Bottom Entry	Confirmed	1
Drop Boxes	Yes/No	Yes, with side panels to allow single person glanding of cables from below (part of switchroom contract)	Drop boxes as part of the switchroom. Side plates are not offered but split gland plates can be provided if required	1
Gland Plates	-	5mm Aluminium, earthed with neoprene gasket	Confirmed	1
Lifting Lugs	-	VTA	Lifting from plinth only	2
Jacking Beam Lugs	-	No – Base of switchboard must be vermin proof and IP rated.	Switchroom gland plates form the seal and are vermin proof	1
Busbars				
Phase Busbar Size	mm^2	VTA	3 x 150 x 10	2
Neutral Busbar Size	mm^2	VTA	3 x 150 x 10	2
Earth Busbar Size	mm^2	VTA	1 x 60 x 6	2

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Phase Dropper Size	mm^2	VTA	1 x 60 x 6 as minimum	2
Neutral Dropper Size	mm^2	VTA	1 x 60 x 6 as minimum	2
Busbar Identification	-	Colour Coded (Red, White, Blue Phases, Black Neutral & Green/Yellow Earth)	Confirmed	1
Busbar Material	-	HDHC Copper	Confirmed	1
Fully Insulated Bus Bars and Droppers	Yes / No	Yes	Confirmed droppers fully segregated	1
Main Busbar Location / Accessibility	-	Top - Front accessible IP6X metal clad segregation	Top. IP4x	3
Phase and Wire Colours				
Phase Colours	-	Red, White, Blue	Confirmed	1
Neutral Colour	-	Black	Confirmed	1
Earth Colour	-	Green / Yellow	Confirmed	1
24V DC +24V DC (+ve)	-	Orange	Confirmed	1
24V DC 0V (-ve)	-	Purple	Confirmed	1
48V DC Tripping and closing (+ve) – HV Swgr	-	Grey	Confirmed	1
48V DC Tripping and closing (-ve) – HV Swgr	-	Black with white tracing	Confirmed	1
Analog 4-20mA	-	Pink	Confirmed	1
Ethernet cables	-	Grey Sheath	Confirmed	1
Fibre optic patch leads	-	Yellow – Single Mode	Confirmed	1
Live Line Indication				
Manufacturer	-	Nilsen	Confirmed	3
Model	-	NWA2019	Confirmed. Alternative options offered	3
Incomer Cubicle	-	Yes – Indicator mounted in control compartment	Confirmed	1
Incomer Cubicle Live Line Sensing	-	Pick up from line side of ACB, as per schematic	Confirmed	1
Outgoing Modules	-	Yes – Indicator mounted in module door. Pick up from downstream side of MCCB, as per schematics	Confirmed	1
Locking				
Incomer padlockable in withdrawn position	Yes / No	Yes - 6.3mm shackle	Confirmed	1
Outgoing MCCB come with mechanically interlocked cubicle door mounted rotary handles so that: * The door cannot be opened in the on position. * The circuit breaker rotary handle can be locked in the off position and the door cannot be opened while padlocked off. * The circuit breaker extension handle shall be lockable inside the module allowing the MCCB to be locked out while the door is open. * The circuit breaker rotary handle cannot be padlocked in the on position.	Yes / No	Yes - 6.3mm shackle	Confirmed	1
Incomer ACB (Rating >1000 Amp)				
Incomer Type	-	3P Withdrawable ACB, padlockable in withdrawn position	3P as per tender clarification	1
Fixed / Demountable / Withdrawable	-	Withdrawable (motorised)	Confirmed	1
Incomer ACB Make	-	NHP Terasaki	Confirmed	1
Incomer ACB Model	-	VTA	AR40	2
Shunt and Trip Coils	-	Yes 24 VDC	Confirmed	1
Undervoltage Trip		Yes 24 VDC	Confirmed	1
Motor Operated Spring Charge	-	Yes 240 VAC	Confirmed	1
External Neutral CT (installed in MCC)	-	Yes	Confirmed	3, 4
Motorised Racking	-	Yes for remote racking utilising TemRack iR	Confirmed	3

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Electronic Trip Unit	-	Yes – AGR-31C Adjustable phase overload time delay curve, definite time and instantaneous protection. Independent Neutral protection. Ground fault Protection. Integral Metering.	Confirmed	3, 4
ACB Communications	-	Via Modbus TCP Communication module. Capable to remote Open/Close via Modbus TCP.	Confirmed	1
Incomer ACB Status Contacts	-	Circuit breaker open (2 off) Circuit breaker closed (2 off) Circuit breaker tripped (1 off) Earth Fault Trip (1 off)	Confirmed. Earth Leakage relay included as per incomer schematic	1
Auxiliary Supply	-	Yes 24 VDC. Trip unit shall be self-powered and auxiliary 24 VDC powered.	Confirmed	1
Remote Control Panel	-	Vendor to make allowance for IO to and from NHP's TemRack IR Local Control Station.	Confirmed	1
Incomer MCCB (Rating up to 1000 Amp)				
Incomer Type	-	Moulded Case Circuit Breaker (Rating up to 1000A) with Earth Leakage	Confirmed. As per Schematics	1
Fixed / Demountable / Withdrawable	-	Fixed	Confirmed	1
Incomer MCCB Make		Terasaki	Confirmed	2
Incomer MCCB Model	-	Tembreak Pro B Models	Confirmed	2
Shunt and Trip Coils	-	Yes 24 VDC	Confirmed	1
Undervoltage Trip		No	Confirmed	1
Motor Operated Spring Charge	-	Yes 240 VAC	Confirmed	1
External Neutral CT (installed in MCC)	-	Yes	Confirmed	3, 4
Motorised Switching	-	Yes, for remote switching	Confirmed	3
MCCB Protection	-	Adjustable Thermal / Magnetic as standard, 3 Pole with earth fault protection, adjustable earth leakage and electronic trip units where shown on SLD. VTA range of adjustment offered for thermal/long time trip.	Confirmed. As per Schematics	3
Incomer MCCB Status Contacts	-	Circuit breaker open (1 off) Circuit breaker closed (1 off) Circuit breaker tripped (2 off)	Confirmed	1
Remote Control Panel	-	Vendor to Design, Supply and install a Pushbutton Local Control Station.	Confirmed	1
Earth Fault Relay (Incomer and Loads)				
Manufacturer	-	VTA - Refer Note 4.	As per tender drawings and clarification	2, 4, 7
Model	-	VTA - Refer Note 4.	Amp control ELV and NHP RD1EP21H	2, 4, 7
Surge Diverters				
Surge Diverters required on Incomer	Yes/No	Yes	Included as per Schematic and SLD	1
Manufacturer	-	VTA	Novaris	2
Model	-	VTA	SD3-200-275	2
Minimum and Maximum Fuse Size	-	VTA	63A	2
Provides Surge Diverter Status Contact to DCS	Yes / No	Yes	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Current Transformers				
Current Transformers – ACB line side				
Manufacturer	-	VTA	Stemar/Ampcontrol	2
Type	Block / Toroid	Toroid	Confirmed	1
Duty	-	Protection Class CT – Arc Detection System Metering Class CT – Power Quality Meter	Confirmed	1
Ratio	-	Refer to Busbar Ratings in Table above	Confirmed	1
Class	-	5P20 5VA (Protection) Cl 1.0 5VA (Metering)	Confirmed	1
Active Harmonic Filter Current Transformers				
Manufacturer	-	VTA	Noted	2
Type	Block / Toroid	Toroid	Confirmed	1
Duty	-	Active harmonic Filter	Confirmed	1
Ratio	-	As per manufacturer requirements.	Noted	2
Class	-	As per manufacturer requirements.	Noted	2
MEN Link Current Transformer				
Manufacturer	-	VTA	Stemar	2
Type	Block / Toroid	Toroid	Confirmed	1
Duty	-	Earth Leakage Protection via external Earth Leakage Relay	Confirmed	1
Ratio	-	As per manufacturer requirements.	Confirmed	2
Class	-	As per manufacturer requirements.	Confirmed	2
Power Quality Meter				
Manufacturer	-	SEL	Confirmed	3
Model		735#824G	Confirmed	
Power Supply	-	240 VAC	Confirmed	1
Communications	-	Via Modbus TCP Communication module	Confirmed	1
Arc Fault Detection System				
Manufacturer	-	Schneider	Confirmed	3
Model	-	VAMP321	V321-BBAAD-AAANA-A1	3
Power Supply	-	24Vdc	Confirmed	1
Communications	-	Via Modbus TCP Communication module	Confirmed	1
Design note	-	Supplier to design, product selection, install and commission Arc fault Detection system come complete with min of 2x fast acting trip contacts rated for a minimum of 48Vdc operating voltage. Current Supervision required for all incomers.	Confirmed. Arc relay settings by others	3, 7
Sensor Locations	-	Incomer and all outgoing modules including cable ways. Arc fault monitoring are not required for areas (E.g. busbar) where the component design is classified as an arc free zone in accordance to IEC61641.	Cable ways and bus zone do not have arc sensors	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Current Transformer	-	As per manufacturer requirements. To be installed by Switchroom Vendor in the Transformer room.	Confirmed	1
VSD Feeders				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
MCCB Make	-	VTA	Terasaki	2
MCCB Model (series)	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Magnetic as standard, rating as per manufacturer's protection table	Confirmed	1
Outgoing cable connection (cable(s) exiting MCC)	-	VTA - Onto MCCB or terminals as per manufacturer's recommendation	Confirmed	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	Confirmed	1
VSD Starter Control Voltage	-	24V DC	Confirmed	1
DOL Motor Starters				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
MCCB Make	-	VTA	Terasaki	2
MCCB Model (series)	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Magnetic as standard, rating as per manufacturer's Type 2 coordination chart for IE3 high efficiency motors. 3 Pole with Solid Neutral sensing, adjustable earth leakage and electronic trip units as shown on Schematics.	Standard 3 pole Thermal Magnetic circuit breaker offered. Earth Leakage provided by NHP relay	1
Contactor Model	-	Type 2 Coordinated AC3 duty suitable for IE3 high efficiency motors	Confirmed	3
Electronic Overload Relay (EOL)				
Electronic Overload Make / Model	-	Rockwell E300 series	Confirmed	1
Electronic Overload functions		As a minimum: - Current sensing - Class 10 motor thermal protection - I/O modules (If required) - PTC Thermistor input	Confirmed. Model included does not have Earth Leakage or Ground Fault protection as this is provided by the Earth Leakage relay	3
Electronic Overload Communication		Modbus TCP – Anybus-E300-MBTCP Comms Module	Confirmed	1
Electronic Overload Expansion I/O module		Where required	As per Schematics	1
Sensing module		Integral or External to suit motor duty	Confirmed	1
HMI/ Starter Control station module		Required	Confirmed	1, 7
Thermal Overload Model (applicable to multi-motor DOL Starters)	-	VTA - Selected to suit Type 2 coordination tables for AC3 duty for high efficiency motors	Confirmed	3
Outgoing cable connection	-	Onto Contactor, terminals, or TOL as per manufacturer's design drawings	Confirmed	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Motor Starter Contactor Coil Voltage	-	240V AC	Confirmed	1
Motor Starter Control Voltage	-	24V DC	Confirmed	1
Soft Starter Motor Starters				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB)	N/A	1
Fixed / Demountable / Withdrawable	-	Demountable	N/A	1
CFS Make	-	VTA	N/A	2
CFS Model	-	VTA	N/A	2
Fuse Type/curve	-	HRC, gR curve, 500V AC, Combined Indicator, Insulated grip lugs.	N/A	1
Contactor Model	-	VTA - Type 2 Coordinated AC3 duty suitable for high efficiency motors	N/A	2
Soft Starter Make / Model	-	VTA - with Modbus TCP Communication module, HMI mounted on MCC cabinet door.	N/A	3
Soft Starter Location	-	Inside MCC cubicle.	N/A	1
Outgoing cable connection	-	Onto soft starter terminals as per design drawings	N/A	1
Neutral Connection	-	Provided inside feeder module via Neutral busbar dropper	N/A	1
Contactor Coil Voltage	-	240V AC	N/A	1
Soft Starter Control Voltage	-	24V DC	N/A	1
MCCB Feeders				
Feeder Type	-	Moulded Case Circuit Breaker (MCCB) with Earth Leakage	Confirmed	1
Fixed / Demountable / Withdrawable	-	Demountable	Confirmed	1
Feeder Make		VTA	Terasaki	2
Feeder MCCB Model	-	VTA	Tembreak Pro	2
MCCB Protection	-	Adjustable Thermal / Magnetic as standard, 3 Pole with Solid Neutral sensing, adjustable earth leakage and electronic trip units where shown on SLD. VTA range of adjustment offered for thermal/long time trip.	Standard 3 pole Thermal Magnetic circuit breaker offered. Earth Leakage provided by NHP relay	3
Fault limiting fuses	-	Yes, where specified on standard drawing.	Confirmed	1
Outgoing cable connection	-	Onto terminals as per manufacturer's design drawings	Confirmed	1
Neutral Connection		Provided inside feeder module via Neutral busbar dropper	Confirmed	1
Standard Motor Starter and VSD Feeder Sizes				
DOL Motor starters, VSD feeders and soft starter motor starters grouped into standard sizes for each starter type with all starters of the same type and size using the same components	Yes / No	Yes	Confirmed	1, 5
Nominated standard sizes	-	Vendor to advise standard module sizes, EPC standard schematics will be adjusted to suit	TBA	3, 5
Control Power Supply & Distribution				
24V DC distribution	-	Redundant 240VAC/24VDC Power supplies fed from Switchroom 240VAC UPS DB.	Confirmed. Power Supply is external to the MCC	1
Separate 24V DC circuit breakers for each tier	Yes / No	Yes – Located at top of cable zone	Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Inter-Panel Wiring				
Routing	-	Routed in cable duct running along top of MCC tiers.	Confirmed	3
Segregation	-	Ethernet Comms to be segregated by metallic barrier or in separate duct.	Confirmed	3
Duct Space for Future Expansion	-	Minimum 20% space to be provided in control wiring ducts for connection of future MCC modules.	Confirmed	1
Communication				
MCC Network Switch (free issued)		Installed in the top of tier cable zones. Maximum of 1 per every two tiers.	Confirmed	2
Labelling				
Switchboard Label	Etched traffolyte label, located on incomer tier, with MCC number, MCC description and SLD number at minimum.		Confirmed	1
Switchboard Nameplate Label	Etched traffolyte nameplate label located on incomer tier. Label shall include the switchboard ratings, serial number, and date of manufacture.		Confirmed	1
Tier Label	Etched traffolyte label (black on white background) to be located on the top of each tier, with the MCC tier number.		Confirmed	1
MCC Module Outer Door Label	Etched traffolyte label (black on white background) on the outer door of each MCC module. Each label shall contain the module tier position, drive kW rating or feeder Amp rating, equipment tag number, equipment description and the MCC schematic drawing number.		Confirmed	1
MCC Module Inner Labelling	Etched traffolyte label (black on white background) with equipment number for equipment internal to MCC module (e.g. MCCB, MCB, Relays, Terminal Strips).		Confirmed	1
Terminals and Panel Wiring	Terminals to be numbered per design drawings, internal wiring shall be marked at both ends as per design drawings.		Confirmed	1
Misc Components				
Miniature Circuit Breakers Make & Model	-	VTA	NHP DIN. 10kA Type	2
MCCB Handles	Yes / No	Yes – Interlocked with cubicle doors. Padlockable both inside and outside of module door.	Confirmed	1
Control Relays	-	VTA Make/Model DC Relays come with integral surge suppression diodes	NHP Finder Type	2
Control and Power Terminal Make & Model	-	Weidmuller WDU series	Confirmed	3
Earth Terminal Make and Model	-	Weidmuller WPE series	Confirmed	3
LED Indicator Lamps & Pushbuttons	-	IPD P9 series	IPD or NHP	3
Voltage test terminals (MCC incomer cubicle)	-	Weidmuller WTL series	Confirmed	3
Earth Leakage Relay – VSD Feeder	Make / Model	AMPCONTROL / ELV	Confirmed	1, 7
Earth Leakage Relay – All other motor starter/ Feeder	Make / Model	NHP / RD1EP21H	Confirmed	1, 7
Test Requirements				
Type (Verification)	**Required,** Provide test certificates		Confirmed	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Routine	**Required,** This includes: - Insulation Test of MCC bus - Functional checks of all MCC modules - ACB Incomer function testing - Primary & Secondary injection testing of AHF and MEN CTs - Primary injection testing of 132+ kW DOL external CTs		Confirmed	1
Shipping and Delivery				
Packaging	Installed in switchroom			1
Dimensions and Weight	Vendor to complete **Table A**			2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. ACB trip unit shall monitor Neutral-Earth Link CT and provide earth fault protection that shall trip ACB and intertrip upstream HV feeder circuit breaker. Where the ACB trip unit is unable to provide said protection, the Supplier shall specify a separate earth fault relay to provide this protection. Where provided, external Earth Fault Relay shall be door mounted to access test/reset buttons, dip-switch settings and view LED indication.
5. Vendor to nominate standard module sizes.
6. Vendor to provide at tender stage an initial AS61439.1 Appendix D.1 Statement of Compliance (SoC) document for the complete assembly with final SoC provided after testing.
7. Where provided, EOL HMI, Arc flash protection relay, power meter, Earth Fault Relays or Earth Leakage Relays shall be door mounted to access push buttons, dip-switch settings and to view indications.

Table A LV MCC Dimensions And Weight

Vendor to complete below table.

NO.	EQUIPMENT NUMBER	LENGTH (mm)	WIDTH (mm)	HEIGHT (mm)	SHIPPING SECTION DIMENSIONS L x H x D (mm)	HEAVIEST SHIPPING SECTION WEIGHT (kg)
1	F12-SW-001	5320	1000	2329		
2	F13-SW-002	6720	1000	2329		
3	F13-SW-003	8580	1000	2329		
4	F13-SW-014	6120	1000	2329		
5	F13-SW-020	7380	1000	2329		
6	F16-SW-004	6520	1000	2329		
7	F16-SW-005	4320	1000	2329		
8	F16-SW-008	1660	700	2329		
9	F21-SW-006	4120	1000	2329		
10	F21-SW-007	4920	1000	2329		
11	F22-SW-030	6120	1000	2329		
12	F22-SW-031	7780	1000	2329		
13	F23-SW-023	4520	1000	2329		
14	F28-SW-009	5320	1000	2329		
15	F72-SW-072					